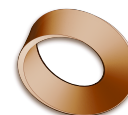


L1 – Modelling & Diagramming (the Pendulum)

Unit 1: Oscillations & Waves • MYP Sciences Year 9

Physics 9A and 9B • Mr. Rawson



9B	Thu 27 Aug, P4 — taught	Source	this plan transcribes that board
9A	Mon 31 Aug, P1, 08:55–09:50	Next	Thu 3 Sep, P6 — 3 days
Criteria	ATL Communication (explicit in the unit planner) groundwork for A language and the B/C investigation — <i>formative, nothing assessed</i>	Lesson	1 of the taught sequence
To print	None — deliberate. Students work in their own notebooks. Do not create a sheet for this lesson.		
Notebooks	The whole lesson happens on a blank notebook page. [[ALEX: decide — whether 9A's notebooks are already issued; if not, issue at the door before anything else]]		
Kit	Whiteboard, red and blue pens. [[ALEX: decide — whether 9B got a real demo pendulum (string + bob) or the board drawing alone; match it for 9A]]		
Prior	9A have had one lesson: course expectations, 27 Aug. Nothing else. This is their first physics content at St Leonards.		
Timing	Built to 9A's Monday P1 — a true 55 (08:55–09:50, bell schedule). 9B's Thursday P4 was a 50; the extra five belongs in the notebook block, not in more content.		

By the end of the lesson students can:

- explain that physicists communicate through **diagrams** (a picture that carries information) and **models** (a deliberate simplification of reality);
- set out a labelled diagram of a swinging pendulum on a blank page — fixed point, arc, bob positions, labels — with no scaffold given;
- state that gravity on Earth is a **vector** that does not change: not magnitude, not direction — so every gravity arrow on the diagram is the same length and points straight down, whatever the mass;
- name one thing the model deliberately leaves out (air resistance) and say why a model is allowed to do that.

Timing — built to 9A at P1 (55 min)

Phases follow the 9B board. [[ALEX: decide — adjust the splits against how the 27 Aug delivery actually ran; the board survives, the clock did not]]

Time	Phase	What happens
0–5	Settle + frame	Board title: <i>Topic 1: Oscillations and Waves</i> . Three boxed headings — Pendulums , Diagramming , Modelling — and Communication underlined in blue. The claim of the day, said plainly: physics is not just knowing things, it is <i>communicating</i> them, and physicists communicate with diagrams and models. Today they learn to do both.
5–13	Terms harvest	The pendulum is the concrete system: something swinging from a fixed point. Class calls out what matters about it; list the terms on the board as they arrive: Motion • Swing • Speed/Velocity • Air Resistance • Distance • Mass • Fixed Point • Gravity . When air resistance comes up, box it in red with “ <i>or not? No atmosphere</i> ” and park it — it pays off in the modelling phase.
13–21	Build the diagram, on the board	Narrate the order so the order itself is the teaching: fixed point at the top first, then the arc of the swing, then the bob drawn at several positions along the arc, then labels (“mass” with an arrow to the bob). A diagram is built, not drawn — each mark carries information.
21–33	Notebooks — their own page	Blank page, no sheet. Each student sets out their own pendulum diagram in the same order and labels it from the term list. Circulate for structure, not neatness: is there a fixed point? an arc? more than one bob position? labels? This block is the reason the lesson exists — protect it, and let the page be theirs.
33–44	The arrows — gravity as a vector	On the board diagram, add a red downward arrow at every bob position. Then the confrontation: two bobs, $m_1 = 2\text{ kg}$ and $m_2 = 4\text{ kg}$ — ask how the arrows should differ. Take predictions first; most expect the heavier bob to get a longer arrow. They do not. Box it in red: <i>Gravity does not change on Earth. Not magnitude, not direction (of arrow)</i> . Name the word: VECTOR — the length of the arrow is the magnitude, the way it points is the direction, and for gravity both are fixed. Write $g = 9.81\text{ m/s}^2$ in red beside the arrows — and write the unit in full: the 9B board read “m/s”, and the squared is worth being precise about, without making a fuss. Students add the arrows, the boxed rule and g to their own diagrams.
44–51	Modelling choices	Return to the red box in the term list: <i>air resistance — or not?</i> A model is a deliberate simplification : we are allowed to imagine a pendulum on a world with no atmosphere if it makes the physics clearer, so long as we say so. That choice — what to keep, what to leave out, and writing the choice down — is what makes it a model rather than a picture. Close the loop to Communication: a good diagram is judged by the information it carries, not by how pretty the pendulum is.
51–55	Close + forward pointer	Today the model came from <i>thinking</i> . Next lesson it comes from <i>measuring</i> : springs, a table, a graph, and a straight line they find themselves. Verbatim close available: “ <i>You have just drawn a model of a system. On Thursday you will build one — from your own data.</i> ”

Questions to ask

1. One bob is 2 kg, the other 4 kg. Why are their gravity arrows the same length? (*The arrow shows gravity itself — 9.81 downwards, the same for every object on Earth. A sharp student may object that the heavier bob is “pulled harder” — that is the force on it, which does grow with mass and gets its own name later. Gravity’s strength does not change, which is why two bobs released together fall together.*)
2. The bob is at the far left of its swing, as high as it gets. Where does its gravity arrow point? (*Straight down. Always. The arrow never tilts with the swing.*)
3. What did our model leave out, and are we allowed to do that? (*Air resistance. Yes — a model is a deliberate simplification, provided the choice is stated.*)
4. What makes your page a *diagram* rather than a drawing? (*Labels, arrows, the fixed point — marks that carry information. Judged by what it tells a reader, not by likeness.*)
5. What could we *measure* about this pendulum? (*Harvest, do not answer: swing time, distance, mass, length... this is the seed of the unit’s pendulum investigation.*)

Common misconceptions

- **“The heavier bob gets a longer arrow.”** The misconception the lesson exists to confront. Take the wrong predictions out loud before correcting — the surprise is the teaching.
- **The arrow tilts with the swing** — drawn along the string, or along the direction of motion. Gravity points straight down at every position on the arc.
- **“Gravity switches off at the top of the swing”** — because the bob is momentarily still. Being still is not the same as being left alone.
- **Speed and velocity used interchangeably.** Both are on the term list; flag that velocity carries a direction — it is the other vector hiding in this lesson. Do not develop it further today.
- **Artwork over information.** Some will produce a beautiful pendulum with no labels. Praise the structure of the plainest correct diagram in the room, by name.

Differentiation

Support: narrate the build order again at the desk — fixed point, arc, bob positions, labels — and let them copy the order, not the board; the red boxed rule can be copied verbatim.

Extend: add a *velocity* arrow to the bob at the bottom of the swing and ask how it differs from the gravity arrow. (*It changes — size and direction — as the bob swings; gravity’s does not.*) Sharpest students: both bobs are released together from the same height — which reaches the bottom first? Defend the answer from the diagram alone.

After the lesson

- **If 9A moves quickly, begin L02 (Hooke’s Law)** — the plan and handout are built. [[ALEX: decide — whether the spring kit can be staged in PH1 for Monday P1 so the extension is actually available]]
- **The L02 paperwork is now stale for 9A:** its plan and note date 9A’s Hooke’s Law delivery Mon 31 Aug P1 — this lesson takes that slot, so 9A’s L02 moves to **Thu 3 Sep, P6**. Update the L02 header table and note, and re-check the “next lesson” gaps.

before Monday]

- Note how far the notebook diagrams got and which students needed the build order re-narrated — that is the support list for the pendulum investigation.

- Record whether notebooks were issued and to whom, if that happened today.
- Flip the Physics 9A Class Log entry to **Taught**.